Lesson 4.2 **Duration: 40 minutes**

Number Systems

Learning Objectives

- Understand the decimal (base-10) number system
- Introduction to binary (base-2) number system
- Compare how different number systems represent values

Engage (5-7 minutes)

Think of counting on fingers versus using an abacus—both represent the same numbers differently. Computers use binary like a abacus with only two positions: on and off. Today we explore how numbers work in different systems.

Explore (10-12 minutes)

Write the numbers 1 to 10 in normal decimal. Then try to write them using only 0s and 1s—binary. Work with a partner to find patterns. How many binary digits do you need to represent the number 5?

Explain (10-12 minutes)

Decimal uses ten digits (0-9) and is base-10 because we have ten fingers. Binary uses only two digits (0 and 1) and is base-2 because computers have two states—on and off. Each position in binary represents a power of 2, just as each decimal position represents a power of 10.

Elaborate (8-10 minutes)

Convert your roll number to binary. For example, roll number 13 becomes 1101 in binary because 8+4+0+1=13. Write a table showing decimal 1-8 and their binary equivalents. Look for the pattern in how binary counts.

Evaluate (5-8 minutes)

Exit ticket: Convert decimal 7 to binary. Explain why computers use binary instead of decimal.